Name- Bhakare Purva Shivaji

Roll no- 08

Practical No- 03

Greedy Best First Search

```
import heapq
graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('E', 1)],
    'D': [('F', 3)],
    'E': [('F', 2)],
    'F': []
}
h = {'A': 7, 'B': 6, 'C': 4, 'D': 3, 'E': 2, 'F': 0}
def greedy(start, goal):
    pq = [(h[start], start, [start])]
    visited = set()
    while pq:
        _, node, path = heapq.heappop(pq)
        if node in visited:
            continue
        visited.add(node)
        if node == goal:
            return path
        for next_node, cost in graph[node]:
            if next_node not in visited:
                heapq.heappush(pq, (h[next_node], next_node, path + [next_node]))
    return None
path = greedy('A', 'F')
print("Greedy Best First Path:", path)
```

OUTPUT-

Greedy Best First Path: ['A', 'C', 'E', 'F']

A* Search Algorithm

```
import heapq
graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('E', 1)],
    'D': [('F', 3)],
    'E': [('F', 2)],
    'F': []
}
h = {'A': 7, 'B': 6, 'C': 4, 'D': 3, 'E': 2, 'F': 0}
def astar(start, goal):
    pq = [(h[start], 0, start, [start])]
    visited = set()
    while pq:
        f, cost, node, path = heapq.heappop(pq)
        if node in visited:
            continue
        visited.add(node)
        if node == goal:
            return path, cost
        for next_node, edge_cost in graph[node]:
            new_cost = cost + edge_cost
            new_f = new_cost + h[next_node]
            if next_node not in visited:
                heapq.heappush(
                    pq, (new_f, new_cost, next_node, path + [next_node])
                )
```

```
    return None, 0
path, cost = astar('A', 'F')
print("A* Path:", path)
print("Path Cost:", cost)
```

OUTPUT-

```
A* Path: ['A', 'B', 'D', 'F']
Path Cost: 6
```